Kamireddy Surendranath Reddy

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Profile Summary

Full Stack Developer and Generative AI engineer who builds systems that work end to end, from React and FastAPI to LLM-powered agents. CSE undergraduate with hands-on experience shipping web applications and AI projects, plus a growing interest in system design. Driven by problem-solving, quality-focused testing, and building scalable, production-ready AI solutions.

Education

B.Tech in Computer Science and Engineering 2022–2026
GITAM University, Bengaluru CGPA: 7.2/10

Intermediate (MPC) 2020–2022
Akanksha Junior College 83.8%

Technical Skills

- **Programming languages:** Java, Python, SQL, TypeScript
- **Web Technologies:** HTML, CSS, React, TanStack (Query, Router)
- **Backend:** FastAPI, SQLAlchemy, Alembic, REST APIs
- **Database:** MySQL, PostgreSQL
- **Generative AI:** LLMs, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Chunking Strategies, Embeddings
- **AI Concepts:** Deep Learning Basics, Image Processing
- **Core Subjects:** Data Structures, Operating Systems, DBMS
- **Tools:** Git, GitHub, VS Code, Antigravity, IntelliJ IDEA, JIRA (Basic)
- **Soft Skills:** Problem Solving, Self-learning, Teamwork, Adaptability

Experience

Genworx.ai – Full Stack & Generative AI Trainee June 2026–October 2026

- Built full stack applications with FastAPI, React, TypeScript, and PostgreSQL, including database migrations using SQLAlchemy and Alembic.
- Developed LLM-powered solutions using RAG, embeddings, chunking strategies, LangChain, and LangGraph.
- Delivered real-world use cases across finance, agriculture, and healthcare, turning business problems into working solutions.
- Collaborated with the team using Git and GitHub, following clean coding and testing practices.

Projects

iClinic – AI Front Desk Assistant (Python, FastAPI, LangGraph, PostgreSQL, React, TypeScript)
GitHub Repository

- Built an AI front desk assistant that books, reschedules, and cancels patient appointments across chat, voice, and outbound phone calls.
- Grounded the agent in each clinic's own data (doctors, branches, visit types, and live free slots) and verified patient identity before taking any action.
- Implemented slot holding with safe, persistent booking to prevent double-bookings, followed by automatic email confirmation to the patient.
- Shipped a login-protected staff dashboard with full conversation history and human hand-over visibility.

Deep Learning-Based Digital Image Watermarking (Python, Deep Learning, CNN, Image Processing)
GitHub Repository

- Developed a deep learning–based encoder–decoder neural network to securely embed and extract digital watermarks from images.
- Designed the model to preserve high visual quality while ensuring robustness against compression, resizing, and noise distortions.
- Evaluated model performance using image quality metrics such as PSNR and SSIM to measure similarity between original and watermarked images.
- Improved digital copyright protection by implementing a reliable and scalable watermarking solution using modern machine learning techniques.

Farma Bridge (React.js, TypeScript, FastAPI, Python, PostgreSQL/SQLite, Alembic)
GitHub Repository

- Developed an agricultural marketplace connecting farmers, buyers, storage facilities, and machinery providers.
- Built a React.js frontend and FastAPI backend with database integration for managing users and agricultural marketplace data.
- **Future Scope**: Integrate an AI-powered crop disease detection assistant to analyze crop images, identify potential diseases, and recommend treatment/prevention measures.

Certifications

- Data Analytics – Cisco Networking Academy (2025)
- AI-ML Virtual Internship – AWS (2023)
- C Programming – Coursera (2022)

Achievements

- Winner – College Cricket Tournament representing GITAM University cricket team.
- Active member of the college cricket team and participated in multiple campus-level sports competitions.
- Participated in South Zone Handball Tournament – CBSE Clusters (2019–2020).

Interests

- Playing Cricket and Volleyball
- Traveling and Exploring New Places